

1. What are the possible orders for the elements of S_6 and A_6 ?
2. Suppose α is a 6-cycle and β is a 5-cycle. Determine whether $\alpha^5\beta^4\alpha^{-1}\beta^{-3}\alpha^5$ is even or odd.
3. Let $\alpha = (1, 3, 5, 7, 9)(2, 4, 6)(8, 10)$. If α^m is a 5-cycle, what can you say about m ?
4. Prove that (1234) is not the product of 3-cycles.
5. Let $\alpha_1, \alpha_2, \alpha_3$ be 2-cycles. Prove that $\alpha_1\alpha_2\alpha_3 \neq \epsilon$.

6. What are the orders of the elements of D_{15} ? How many elements have these orders?
7. Suppose G is a group with exactly eight elements of order 10. How many cyclic subgroups of order 10 does G have?
8. Prove that a group of order 4 must be Abelian.
9. The *torsion subgroup* is the subset of elements of finite order in an Abelian group G . Prove that this is a subgroup of G .

10. Suppose G is a group and G has exactly two non-trivial proper subgroups. Prove that G is cyclic and $|G| = pq$, where p and q are distinct primes, or that G is cyclic and $|G| = p^3$, where p is a prime.

11. Let G be a cyclic group of order n and let H be the subgroup of G of order d . Show that $H = \{x \in G : |x| \text{ divides } d\}$.